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Advanced Methods for Scientific Computing:

Fast Multipole Methods for N-Body Problems

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1 | Introduction

The N-body problem arises in many physical applications, including celestial mechanics, plasma physics, molecular dynamics, and fluid dynamics. In its general formulation, the quantity associated with each target particle is obtained by summing the contributions of all source particles. For N particles, the direct evaluation of all pairwise interactions requires $O(N^2)$ computational work, which rapidly becomes prohibitive as the number of particles increases.

The **Fast Multipole Method** (FMM), introduced by Greengard and Rokhlin [1][4], provides an efficient approach to reduce the computational cost of such interactions. Its key idea is to exploit the spatial structure of the problem: particles that are sufficiently far from a target do not need to be considered individually. Instead, their collective effect can be approximated by a compact representation, known as a **multipole expansion**. In this way, the interactions are separated into near-field contributions, which can be computed directly, and far-field contributions, which are evaluated through approximations based on groups of particles.

2 | N-Body Problem

2.1. Problem Formulation

The N-body problem consists in computing the field generated by a set of N source particles. Each particle j is characterized by a position $\mathbf{x}_j$ and a source strength q_j. The physical potential at a target point $\mathbf{y}_i$ is written in the general form

$$\phi(\mathbf{y}_i) = \sum_{j=1}^{N} q_j G(\mathbf{y}_i, \mathbf{x}_j), \tag{2.1}$$

where $G(\mathbf{y}_i, \mathbf{x}_j)$ denotes the interaction kernel. The quantity ϕ represents the physical field to be computed and is real-valued.

In the direct approach, the interaction is evaluated for every pair of source and target particles. The computational cost is therefore $\mathcal{O}(N^2)$, which becomes prohibitive for large systems.

The Fast Multipole Method (FMM) reduces this computational cost by grouping particles according to their spatial location and exploiting the fact that the collective effect of a well-separated group of sources can be approximated by a compact expansion.

Interactions between nearby particles are evaluated directly, whereas interactions between well-separated groups are represented through multipole and local expansions. The resulting field can consequently be decomposed into near- and far-field contributions,

$$\phi_i = \phi_i^{\text{near}} + \phi_i^{\text{far}}. \tag{2.2}$$

The near-field contribution is computed by direct evaluation,

$$\phi_i^{\text{near}} = \sum_{j \in \mathcal{N}(i)} q_j G(\mathbf{y}_i, \mathbf{x}_j), \tag{2.3}$$

where $\mathcal{N}(i)$ denotes the set of source particles belonging to the near field of target particle i.

Conversely, the far-field contribution is approximated by means of multipole and local expansions. Distant groups of source particles are first represented collectively through multipole expansions, which are then translated into local expansions centered at the corresponding target cells. This allows the collective effect of distant particles to be evaluated without computing each particle-to-particle interaction individually. Consequently, the FMM avoids the direct evaluation of all N^2 particle interactions while retaining an accurate treatment of nearby interactions.

3 | Fast Multipole Method

The FMM algorithm organizes the computation of the far-field contribution through a hierarchical representation of the computational domain. The particles are grouped into spatial cells, which are recursively subdivided to form a tree structure.
The resulting hierarchy is then traversed through an **upward pass**, an **interaction phase**, and a **downward pass**.

3.1. Tree construction

The computational domain is recursively subdivided into smaller cells, forming a hierarchical tree. Each cell is associated with a set of particles and a geometric center. The subdivision continues until a prescribed criterion is satisfied, defining the leaf cells of the tree.

At the leaf level, the particles contained in each cell are used to construct a multipole representation of the corresponding source distribution. These representations are subsequently propagated through the hierarchy during the upward pass.

The three phases make easier manipulate and approximate the fields along this tree, avoiding the need to calculate directly every single pair of particles.
To understand them well, we first clarify the difference between Multipole Expansion and Local Expansion:

- **Multipole** (M): describes the field *generated by* a group of sources "viewed from afar" (outgoing field).
- **Local** (L): describes the field *perceived in* a specific region "coming from distant sources" (incoming field).

3.2. Upward Pass: Multipole-to-Multipole

The upward pass starts from the leaf cells and proceeds towards the root of the tree. At each level, the multipole representations of the children are translated to the center of their parent cell and combined into a single multipole representation.

This operation is referred to as the Multipole-to-Multipole (M2M) translation. It allows a group of source particles contained in a parent cell to be represented by a single expansion, independently of the individual particles.

3.3. Interaction Phase: Multipole-to-Local

Once the multipole representations have been constructed, the far-field interactions between source and target cells are evaluated. For each target cell, well-separated source cells are identified through the corresponding interaction list.

The multipole representation of each admissible source cell is then translated into a local representation centered at the target cell. This operation is referred to as the Multipole-to-Local (M2L) translation.

The resulting local representation collects the far-field contribution of a group of well-separated source particles to the target cell, without requiring the individual source-target interactions to be evaluated.

3.4. Downward Pass: Local-to-Local

The local representations obtained during the interaction phase are propagated down the target tree. Starting from a target parent cell, its local expansion is translated to the centers of its child cells. This operation is referred to as the Local-to-Local (L2L) translation.

The M2L and L2L operations are recursively applied as the tree is traversed towards the leaves. At each level, the local representation accumulated at a parent cell is propagated to its children through the L2L translation. This process continues until a leaf cell is reached, at which point the downward pass terminates.

3.5. Evaluation

Once a leaf cell is reached, the M2L translation is performed with the leaf cell as the target cell and the source cells given by its interaction list. The resulting local representation contains the far-field contribution from all well-separated source cells and is evaluated at the positions of the target particles within the leaf, yielding ϕ_i^{far}.

The remaining near-field interactions, which are not included in the multipole approximation, are then computed directly between the target particles and the particles belonging to the corresponding neighboring source leaf cells, using the original interaction kernel. The total potential at each target particle is therefore given by

$$\boxed{\phi_i = \phi_i^{\text{near}} + \phi_i^{\text{far}}}. \tag{3.1}$$

The resulting potential is then used by the particle-dynamics model to determine the evolution of the particle positions.

4 | Mathematical Formulation

This chapter presents the mathematical formulation adopted for the FMM implementation. The interaction kernel and the corresponding multipole and local representations are introduced separately for the two- and three-dimensional cases. For each formulation, the translation operators required by the FMM algorithm are then derived and used to construct the corresponding interaction approximations.

4.1. Two-Dimensional Formulation

In the two-dimensional case, particle positions can be represented using complex numbers. Given a particle position (x, y), we define the corresponding complex coordinate as

$$z = x + iy, \tag{4.1}$$

where i is the imaginary unit. Similarly, the position of the j-th source particle is denoted by z_j, while z denotes a target point.

The interaction between source and target particles is described by a kernel function $G(\mathbf{x}, \mathbf{x}_j)$.

In the present work, we choose the logarithmic kernel

$$G(\mathbf{x}, \mathbf{x}_j) = -\log \|\mathbf{x} - \mathbf{x}_j\| \,. \tag{4.2}$$

Using complex coordinates, the distance between the target and source points can be written as

$$\|\mathbf{x} - \mathbf{x}_j\| = |z - z_j|, \tag{4.3}$$

and therefore

$$G(z, z_j) = -\log |z - z_j|. \tag{4.4}$$

The physical potential generated by all source particles is therefore

$$\phi(z) = -\sum_j q_j \log |z - z_j|. \tag{4.5}$$

For the purpose of the FMM, the interactions are divided into near-field and far-field contributions. The near-field interactions are evaluated directly using the original interaction kernel, while the far-field interactions between sufficiently well-separated cells are approximated using a multipole expansion.

The near-field contribution is given by

$$\boxed{\phi^{\text{near}}(z) = -\sum_{j \in \mathcal{N}(z)} q_j \log |z - z_j|} \tag{4.6}$$

where $\mathcal{N}(z)$ denotes the set of source particles belonging to the near-field cells of the target point z.

The remaining interactions form the far-field contribution,

$$\boxed{\phi^{\text{far}}(z) = -\sum_{j \in \mathcal{F}(z)} q_j \log |z - z_j|} \tag{4.7}$$

where $\mathcal{F}(z)$ denotes the source particles whose interactions with the target are **treated using the multipole approximation**. Hence,

$$\phi(z) = \phi^{\text{near}}(z) + \phi^{\text{far}}(z). \tag{4.8}$$

To derive the multipole representation of the far-field contribution, we introduce the complex-valued auxiliary function

$$U(z) = \sum_{j \in \mathcal{F}(z)} q_j \log(z - z_j), \tag{4.9}$$

whose real part is related to the physical far-field potential by

$$\phi^{\text{far}}(z) = -\operatorname{Re}\left[U(z)\right]. \tag{4.10}$$

Let z_c denote the center of a source cell. For each source particle in this cell, we write

$$z - z_j = (z - z_c) - (z_j - z_c). \tag{4.11}$$

Hence,

$$\log(z - z_j) = \log(z - z_c) + \log\left(1 - \frac{z_j - z_c}{z - z_c}\right). \tag{4.12}$$

For a target point sufficiently far from the source cell, the following condition holds:

$$\left| \frac{z_j - z_c}{z - z_c} \right| < 1. \tag{4.13}$$

The logarithm can therefore be expanded using the Taylor series

$$\log(1-\xi) = -\sum_{k=1}^{\infty} \frac{\xi^k}{k}, \qquad |\xi| < 1. \tag{4.14}$$

Substituting

$$\xi = \frac{z_j - z_c}{z - z_c}, \tag{4.15}$$

gives

$$\log(z - z_j) = \log(z - z_c) - \sum_{k=1}^{\infty} \frac{1}{k} \frac{(z_j - z_c)^k}{(z - z_c)^k}. \tag{4.16}$$

Summing over all source particles contained in the source cell yields

$$U(z) = \left(\sum_j q_j \right) \log(z - z_c) - \sum_{k=1}^{\infty} \frac{1}{k} \frac{\sum_j q_j (z_j - z_c)^k}{(z - z_c)^k}. \tag{4.17}$$

This naturally leads to the definition of the multipole coefficients. The zeroth-order coefficient is

$$M_0 = \sum_j q_j, \tag{4.18}$$

while, for $k \geq 1$,

$$\boxed{M_k = \sum_j q_j (z_j - z_c)^k}. \tag{4.19}$$

The complex multipole expansion can therefore be written as

$$U(z) = M_0 \log(z - z_c) - \sum_{k=1}^{\infty} \frac{M_k}{k(z - z_c)^k}. \tag{4.20}$$

In practice, the expansion is truncated at a finite order p:

$$\boxed{U(z) \approx M_0 \log(z - z_c) - \sum_{k=1}^{p} \frac{M_k}{k(z - z_c)^k}}. \tag{4.21}$$

The corresponding far-field contribution to the physical potential is finally recovered by taking the real part:

$$\boxed{\phi^{\text{far}}(z) \approx -\operatorname{Re}\left[M_0 \log(z - z_c) - \sum_{k=1}^{p} \frac{M_k}{k(z - z_c)^k}\right]}. \tag{4.22}$$

The total physical potential is then obtained by combining the direct near-field contribution with the approximated far-field contribution,

$$\boxed{\phi(z) = \phi^{\text{near}}(z) + \phi^{\text{far}}(z)}. \tag{4.23}$$

4.1.1. Multipole-to-Multipole (M2M)

Consider a parent cell P with center z_P and its four children $C_1, \ldots, C_4$, whose centers are denoted by z_{C_c}. Each child cell has already computed its multipole coefficients

$$M_n^{(C_c)} = \sum_{j \in C_c} q_j (z_j - z_{C_c})^n, \qquad n = 0, \ldots, p. \tag{4.24}$$

The multipole coefficients of the parent cell are defined with respect to the parent center,

$$M_k^{(P)} = \sum_{j \in P} q_j (z_j - z_P)^k. \tag{4.25}$$

Since the parent contains the four child cells, this expression can be written as

$$M_k^{(P)} = \sum_{c=1}^{4} \sum_{j \in C_c} q_j (z_j - z_P)^k. \tag{4.26}$$

For a particle belonging to child C_c, the displacement with respect to the parent center can be decomposed as

$$z_j - z_P = (z_j - z_{C_c}) + (z_{C_c} - z_P). \tag{4.27}$$

Applying the binomial theorem gives

$$(z_j - z_P)^k = \sum_{n=0}^{k} \binom{k}{n} (z_j - z_{C_c})^n (z_{C_c} - z_P)^{k-n}. \tag{4.28}$$

Substituting this expression into the definition of the parent multipole coefficients gives

$$M_k^{(P)} = \sum_{c=1}^{4} \sum_{n=0}^{k} \binom{k}{n} (z_{C_c} - z_P)^{k-n} \sum_{j \in C_c} q_j (z_j - z_{C_c})^n. \tag{4.29}$$

Recognizing the inner sum as the n-th multipole coefficient of the child cell, we obtain the M2M translation formula

$$\boxed{M_k^{(P)} = \sum_{c=1}^{4} \sum_{n=0}^{k} \binom{k}{n} M_n^{(C_c)} (z_{C_c} - z_P)^{k-n}}, \qquad k = 0, \dots, p. \tag{4.30}$$

Therefore, the multipole expansion of the parent cell is obtained by translating and summing the multipole expansions of its four children. The operation can be summarized as

$$\left\{M_n^{(C_1)}\right\}, \dots, \left\{M_n^{(C_4)}\right\} \quad \xrightarrow{\text{M2M}} \quad \left\{M_k^{(P)}\right\}. \tag{4.31}$$

For an expansion truncated at order $p = 2$, the first three parent multipole coefficients are

$$M_0^{(P)} = \sum_{c=1}^{4} M_0^{(C_c)}, \tag{4.32}$$

$$M_1^{(P)} = \sum_{c=1}^{4} \left[M_1^{(C_c)} + M_0^{(C_c)} (z_{C_c} - z_P) \right], \tag{4.33}$$

and

$$M_2^{(P)} = \sum_{c=1}^{4} \left[M_2^{(C_c)} + 2M_1^{(C_c)} (z_{C_c} - z_P) + M_0^{(C_c)} (z_{C_c} - z_P)^2 \right]. \tag{4.34}$$

4.1.2. Multipole-to-Local (M2L)

Interaction List

For a given target cell T at level ℓ, the interaction list $\mathcal{I}(T)$ contains the cells that are sufficiently far from T to be treated through an M2L translation, but whose parent cells are still neighbours of the parent of T. Let $\mathcal{N}(T)$ denote the neighbourhood of T, and let $P(T)$ be the parent of T. The interaction list is therefore defined as

$$\mathcal{I}(T) = \{S \in \text{Children}(\mathcal{N}(P(T))) \; : \; S \notin \mathcal{N}(T)\}. \tag{4.35}$$

Equivalently, $\mathcal{I}(T)$ is obtained by collecting the children of the neighbouring cells of the parent of T and removing all cells that belong to the direct neighbourhood of T. This construction ensures that the source cells in $\mathcal{I}(T)$ are well-separated from T and can therefore be handled by the M2L operator.

Let S denote a source cell with center z_S, and let T denote a well-separated target cell with center z_T. The source cell is represented by the multipole coefficients

$$M_k^{(S)} = \sum_{j\in S} q_j (z_j - z_S)^k, \qquad k = 0, \ldots, p. \tag{4.36}$$

The far-field generated by the source cell can therefore be written as

$$U^{(S)}(z) = M_0^{(S)} \log(z - z_S) - \sum_{k=1}^{p} \frac{M_k^{(S)}}{k(z - z_S)^k}. \tag{4.37}$$

Since the target point z belongs to the target cell T, it is more convenient to express this expansion with respect to the target cell center z_T. We write

$$z - z_S = (z - z_T) + (z_T - z_S). \tag{4.38}$$

Defining the displacement between the two cell centers as

$$d = z_T - z_S, \tag{4.39}$$

we obtain

$$z - z_S = d + (z - z_T). \tag{4.40}$$

Since the source and target cells are well separated, the following condition holds:

$$\left| \frac{z - z_T}{d} \right| < 1. \tag{4.41}$$

The logarithmic term can then be expanded as

$$\log(z - z_S) = \log d + \log\left(1 + \frac{z - z_T}{d}\right), \tag{4.42}$$

where

$$\log(1 + \xi) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \xi^n, \qquad |\xi| < 1. \tag{4.43}$$

Therefore,

$$\log(z - z_S) = \log d + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \frac{(z - z_T)^n}{d^n}. \tag{4.44}$$

Similarly, for $k \geq 1$, the inverse powers can be expanded using the binomial series,

$$\frac{1}{(z - z_S)^k} = \frac{1}{d^k} \left(1 + \frac{z - z_T}{d}\right)^{-k}, \tag{4.45}$$

with

$$(1 + \xi)^{-k} = \sum_{n=0}^{\infty} (-1)^n \binom{k + n - 1}{n} \xi^n. \tag{4.46}$$

Consequently,

$$\frac{1}{(z - z_S)^k} = \sum_{n=0}^{\infty} (-1)^n \binom{k + n - 1}{n} \frac{(z - z_T)^n}{d^{k+n}}. \tag{4.47}$$

Substituting these expansions into the multipole representation gives

$$\begin{aligned} U^{(S)}(z) = M_0^{(S)} \log d + M_0^{(S)} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \frac{(z - z_T)^n}{d^n} \\ - \sum_{k=1}^{p} \frac{M_k^{(S)}}{k} \sum_{n=0}^{\infty} (-1)^n \binom{k + n - 1}{n} \frac{(z - z_T)^n}{d^{k+n}}. \end{aligned} \tag{4.48}$$

Collecting the terms associated with the same power of $(z - z_T)$ leads to a local expansion centered at the target cell,

$$\boxed{U^{(S)}(z) = \sum_{n=0}^{\infty} L_n^{(T,S)} (z - z_T)^n}. \tag{4.49}$$

The zeroth-order local coefficient is

$$\boxed{L_0^{(T,S)} = M_0^{(S)} \log(z_T - z_S) - \sum_{k=1}^{p} \frac{M_k^{(S)}}{k(z_T - z_S)^k}}. \tag{4.50}$$

For $n \geq 1$, the local coefficients are given by

$$\boxed{L_n^{(T,S)} = (-1)^{n+1} \left[\frac{M_0^{(S)}}{n(z_T - z_S)^n} + \sum_{k=1}^{p} \binom{k+n-1}{n} \frac{M_k^{(S)}}{k(z_T - z_S)^{k+n}} \right]}. \tag{4.51}$$

The total local coefficient $L_n^{(T)}$ is then obtained by summing the contributions from all source cells S belonging to the interaction list $\mathcal{I}(T)$ of the target cell T:

$$\boxed{L_n^{(T)} = L_{n,\mathrm{L2L}}^{(T)} + \sum_{S \in \mathcal{I}(T)} L_{n,\mathrm{M2L}}^{(T,S)}} \tag{4.52}$$

where $L_{n,\mathrm{L2L}}^{(T)}$ denotes the contribution propagated from T's parent cell through the L2L operation, explained in the following paragraph.

4.1.3. Local-to-Local (L2L)

Let P denote a target parent cell with center z_P, and let C denote one of its four children, with center z_C. Suppose that the local expansion associated with the parent cell is

$$U^{(P)}(z) = \sum_{n=0}^{p} L_n^{(P)} (z - z_P)^n. \tag{4.53}$$

For a target point z belonging to the child cell C, the displacement with respect to the parent center can be written as

$$z - z_P = (z - z_C) + (z_C - z_P). \tag{4.54}$$

Applying the binomial theorem gives

$$(z - z_P)^n = \sum_{k=0}^{n} \binom{n}{k} (z - z_C)^k (z_C - z_P)^{n-k}. \tag{4.55}$$

Substituting this expression into the local expansion of the parent cell yields

$$U^{(P)}(z) = \sum_{n=0}^{p} L_n^{(P)} \sum_{k=0}^{n} \binom{n}{k} (z - z_C)^k (z_C - z_P)^{n-k}. \tag{4.56}$$

Rearranging the summation by collecting equal powers of $(z - z_C)$, the expansion can be written as

$$\boxed{U^{(C)}(z) = \sum_{k=0}^{p} L_k^{(C)} (z - z_C)^k} \tag{4.57}$$

where the local coefficients of the child are given by

$$\boxed{L_k^{(C)} = \sum_{n=k}^{p} \binom{n}{k} L_n^{(P)} (z_C - z_P)^{n-k}}, \qquad k = 0, \ldots, p. \tag{4.58}$$

Thus, the L2L operation translates the local expansion from the parent center z_P to the center z_C of the child cell.

For an expansion truncated at order $p = 2$, the first three local coefficients are

$$L_0^{(C)} = L_0^{(P)} + L_1^{(P)}(z_C - z_P) + L_2^{(P)}(z_C - z_P)^2, \tag{4.59}$$

$$L_1^{(C)} = L_1^{(P)} + 2L_2^{(P)}(z_C - z_P), \tag{4.60}$$

and

$$L_2^{(C)} = L_2^{(P)}. \tag{4.61}$$

The downward propagation is repeated recursively until the leaf cells are reached, where the local expansion can be evaluated at the target particles.

4.1.4. Evaluation

After the downward pass, each target leaf cell contains a local expansion centered at its cell center z_T. This expansion represents the far-field contribution generated by the well-separated source cells associated with the target cell.

For a target particle located at z_i, the local expansion is evaluated at its position as

$$U_i = \sum_{n=0}^{p} L_n^{(T)} (z_i - z_T)^n. \tag{4.62}$$

Since U is a complex-valued auxiliary function, the corresponding physical far-field potential is obtained by taking its real part,

$$\boxed{\phi_i^{\text{far}} = -\operatorname{Re}\left[\sum_{n=0}^{p} L_n^{(T)} (z_i - z_T)^n\right].} \tag{4.63}$$

The interactions between the target particle and nearby source particles are not included in the local expansion. These near-field interactions are evaluated directly using the original two-dimensional kernel,

$$\phi_i^{\text{near}} = \sum_{j \in \mathcal{N}(i)} q_j \log |z_i - z_j|. \tag{4.64}$$

For the complex coordinates

$$z_i = x_i + \mathrm{i}y_i, \qquad z_j = x_j + \mathrm{i}y_j, \tag{4.65}$$

the modulus of their difference is

$$|z_i - z_j| = |(x_i - x_j) + \mathrm{i}(y_i - y_j)| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \tag{4.66}$$

Therefore, the near-field contribution can equivalently be written entirely in terms of real coordinates as

$$\boxed{\phi_i^{\text{near}} = \sum_{j \in \mathcal{N}(i)} q_j \log \left(\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \right).} \tag{4.67}$$

The total potential at the target particle is finally obtained by combining the near-field and far-field contributions,

$$\boxed{\phi_i = \phi_i^{\text{near}} + \phi_i^{\text{far}}.} \tag{4.68}$$

4.2. Three-Dimensional Formulation

In the three-dimensional case, particle positions are represented by vectors in $\mathbb{R}^3$. Let $\mathbf{x}$ denote a target point and $\mathbf{x}_j$ the position of the j-th source particle.

We consider the three-dimensional kernel

$$\boxed{G(\mathbf{x}, \mathbf{x}_j) = \frac{1}{|\mathbf{x} - \mathbf{x}_j|}}. \tag{4.69}$$

This kernel corresponds, up to a multiplicative normalization constant, to the fundamental solution of the three-dimensional Laplace equation. In the present formulation, the multiplicative constant is absorbed into the definition of the kernel. If required by the physical model, the corresponding physical normalization constant can be reintroduced when computing the resulting potential and particle forces.

The potential generated by a set of source particles with strengths q_j is therefore defined as

$$\phi(\mathbf{x}) = \sum_j q_j G(\mathbf{x}, \mathbf{x}_j), \tag{4.70}$$

or, explicitly,

$$\boxed{\phi(\mathbf{x}) = \sum_j \frac{q_j}{|\mathbf{x} - \mathbf{x}_j|}}. \tag{4.71}$$

As in the two-dimensional formulation, the FMM separates the evaluation of the potential into near-field and far-field contributions. Interactions between nearby source and target particles are evaluated directly, while the contribution of well-separated source cells is approximated through multipole and local expansions.

For the far-field contribution, the source particles contained in a cell are first represented by a multipole expansion centered at the cell center. This multipole representation is subsequently translated to local expansions centered at target cells through the multipole-to-local (M2L) operator. The resulting local expansions are then propagated through the tree using the local-to-local (L2L) translations and finally evaluated at the target particles.

4.2.1. Spherical Harmonics and Translation Operators

In three dimensions, the multipole and local expansions are expressed using spherical harmonics and the corresponding regular and singular solid harmonics. Let a point $\mathbf{x} \in \mathbb{R}^3$ be represented in spherical coordinates as

$$\mathbf{x} = (r, \theta, \varphi), \tag{4.72}$$

where r denotes the radial distance from the expansion center, θ the polar angle, and φ the azimuthal angle.

The spherical harmonics are denoted by

$$\boxed{Y_n^m(\theta, \varphi) = C_n^m \cdot P_n^{|m|}(\cos\theta) \cdot e^{im\varphi}} \qquad n \geq 0, \quad -n \leq m \leq n. \tag{4.73}$$

where:

$$C_n^m = \sqrt{\frac{(n - |m|)!}{(n + |m|)!}}, \tag{4.74}$$

$$P_n^{|m|}(x) = (-1)^{|m|}(1 - x^2)^{|m|/2} \frac{d^{|m|}}{dx^{|m|}} P_n(x) \tag{4.75}$$

where:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n \tag{4.76}$$

From the spherical harmonics, we define the regular solid harmonics as

$$\boxed{R_n^m(\mathbf{x}) = r^n Y_n^m(\theta, \varphi)} \tag{4.77}$$

The stable recursive algorithm for $\hat{R}_n^m$ that we used is:

- **base case**

$$\hat{R}_0^0 = 1, \tag{4.78}$$

- **diagonal step** ($m = n$)

$$\hat{R}_n^n = -(x + iy)\sqrt{\frac{2n-1}{2n}}\,\hat{R}_{n-1}^{n-1}, \quad n = 1, \ldots, p \tag{4.79}$$

- **three-term recurrence** ($n > m \geq 0$, **fixed column**)

$$\hat{R}_n^m = \frac{(2n-1)\,z\,\hat{R}_{n-1}^m - \sqrt{(n-1)^2 - m^2}\,r^2\,\hat{R}_{n-2}^m}{\sqrt{n^2 - m^2}}, \quad n = m+1, \ldots, p \tag{4.80}$$

and the singular solid harmonics as

$$\boxed{S_n^m(\mathbf{x}) = r^{-(n+1)} Y_n^m(\theta, \varphi).} \tag{4.81}$$

The regular solid harmonics are regular at the origin and are used to represent local expansions, whereas the singular solid harmonics are singular at the origin and are used to represent the far-field multipole expansion.

For a cell with center $\mathbf{c}$, the position of a particle is always expressed relative to that center. Thus, for a particle at $\mathbf{x}_j$, its relative position is

$$\mathbf{r}_j = \mathbf{x}_j - \mathbf{c}. \tag{4.82}$$

The multipole coefficients associated with a source cell C can then be expressed in terms of the regular solid harmonics as

$$\boxed{M_n^m(C) = \sum_{j \in C} q_j R_n^{-m}\left(\mathbf{x}_j - \mathbf{c}_C\right).} \tag{4.83}$$

The use of cell-relative coordinates is essential for the FMM, because the same particle distribution can be represented with respect to different expansion centers.

Changing the center of an expansion is performed through translation operators. In the three-dimensional FMM, three translation operators are required:

$$\boxed{T_{MM}, \qquad T_{ML}, \qquad T_{LL}.} \tag{4.84}$$

The multipole-to-multipole operator T_{MM} translates a multipole expansion from a child cell center to its parent center,

$$\mathbf{M}^{(C \to P)} = T_{MM}(\mathbf{d}_{CP})\mathbf{M}^{(C)}. \tag{4.85}$$

The multipole-to-local operator T_{ML} converts a multipole expansion centered at a source cell into a local expansion centered at a well-separated target cell,

$$\mathbf{L}^{(S\to T)} = T_{ML}(\mathbf{d}_{ST})\mathbf{M}^{(S)}. \tag{4.86}$$

Finally, the local-to-local operator T_{LL} translates a local expansion from a parent cell to one of its children,

$$\mathbf{L}^{(P\to C)} = T_{LL}(\mathbf{d}_{PC})\mathbf{L}^{(P)}. \tag{4.87}$$

Here, each displacement vector describes the relative position of the new expansion center with respect to the original one. The precise form of the translation coefficients depends on the chosen normalization of the spherical and solid harmonics.

These operators provide the mathematical basis for the upward and downward passes of the FMM. Their application allows the algorithm to change the center of an expansion without returning to the individual source particles.

4.2.2. Multipole-to-Multipole Translation

Let C be a child cell of a parent cell P, with centers $\mathbf{c}_C$ and $\mathbf{c}_P$, respectively. The multipole expansion associated with the child cell is centered at $\mathbf{c}_C$ and is described by the coefficients

$$M_n^m(C) = \sum_{j\in C} q_j R_n^{-m}\left(\mathbf{x}_j - \mathbf{c}_C\right), \qquad 0 \le n \le p, \quad -n \le m \le n. \tag{4.88}$$

Now suppose that N charges have strengths $\mathbf{q_1}, \mathbf{q_2}, \ldots, \mathbf{q_N}$, the potential due to these charges is given by the multipole expansion in spherical coordinathes $X = (\theta, \phi, r)$:

$$\Phi(X) = \sum_{n=0}^{\infty}\sum_{m=-n}^{n} \frac{M_n^m}{r^{n+1}} \cdot Y_n^m(\theta,\phi) = \sum_{n=0}^{\infty}\sum_{m=-n}^{n} M_n^m \cdot S_n^m(X) \tag{4.89}$$

where

$$M_n^m = \sum_{j=0}^{n}\sum_{k=-j}^{j} \frac{O_{n-j}^{m-k} \cdot i^{|m|-|k|-|m-k|} \cdot A_j^k \cdot A_{n-j}^{m-k} \cdot \rho^j \cdot Y_j^{-k}(\alpha,\beta)}{A_n^m}, \tag{4.90}$$

with A_j^k defined by the formula

$$A_j^k = \frac{(-1)^j}{\sqrt{(j-k)! \cdot (j+k)!}}. \tag{4.91}$$

During the upward pass of the FMM, these coefficients must be translated from the center of the child cell to the center of its parent. Define the displacement vector

$$\boxed{\mathbf{d}_{CP} = \mathbf{c}_C - \mathbf{c}_P}. \tag{4.92}$$

For a source particle $j \in C$, its position relative to the parent center can then be written as

$$\mathbf{x}_j - \mathbf{c}_P = (\mathbf{x}_j - \mathbf{c}_C) + \mathbf{d}_{CP}. \tag{4.93}$$

The translation of the multipole expansion is based on the addition theorem for the regular solid harmonics. In general, a translated regular solid harmonic can be expressed as a linear combination of regular solid harmonics centered at the original point,

$$R_n^{-m}(\mathbf{r} + \mathbf{d}_{CP}) = \sum_{k,l} T_{n,m}^{k,l}(\mathbf{d}_{CP})\, R_k^{-l}(\mathbf{r}), \tag{4.94}$$

where $T_{n,m}^{k,l}(\mathbf{d}_{CP})$ are the coefficients of the multipole-to-multipole translation operator.

Applying this relation to the multipole coefficients of the child gives

$$M_n^m(C \to P) = \sum_{k,l} T_{n,m}^{k,l}(\mathbf{d}_{CP})\, M_k^l(C). \tag{4.95}$$

Equivalently, collecting all multipole coefficients into a vector, the M2M operation can be written in compact form as

$$\boxed{\mathbf{M}^{(C\to P)} = T_{MM}(\mathbf{d}_{CP})\, \mathbf{M}^{(C)}.} \tag{4.96}$$

The parent multipole coefficients are then obtained by summing the translated contributions from all its children:

$$\boxed{\mathbf{M}^{(P)} = \sum_{C \in \mathcal{C}(P)} T_{MM}(\mathbf{c}_C - \mathbf{c}_P)\, \mathbf{M}^{(C)},} \tag{4.97}$$

where $\mathcal{C}(P)$ denotes the set of children of the parent cell P.

Thus, the M2M operation does not require the individual source particles to be revisited. The multipole representation already computed for each child is translated to the parent center and then combined with the contributions of the other children.

4.2.3. Multipole-to-Local Translation

During the downward pass of the FMM, the far-field contribution of well-separated source cells is transferred to local expansions centered at the corresponding target cells. Let S denote a source cell and T a target cell belonging to the interaction list of T, with centers $\mathbf{c}_S$ and $\mathbf{c}_T$, respectively.

The multipole expansion associated with the source cell is centered at $\mathbf{c}_S$ and is represented by the coefficients $M_n^m(S)$. The displacement between the two expansion centers is defined as

$$\boxed{\mathbf{d}_{ST} = \mathbf{c}_T - \mathbf{c}_S}. \tag{4.98}$$

For a target point $\mathbf{x}$, its position relative to the source center can therefore be written as

$$\mathbf{x} - \mathbf{c}_S = (\mathbf{x} - \mathbf{c}_T) + \mathbf{d}_{ST}. \tag{4.99}$$

The multipole-to-local translation theorem allows the multipole expansion centered at $\mathbf{c}_S$ to be re-expressed as a local expansion centered at $\mathbf{c}_T$. In terms of the translation operator, this conversion is written as

$$\boxed{\mathbf{L}^{(S\to T)} = T_{ML}\left(\mathbf{d}_{ST}\right)\mathbf{M}^{(S)}.} \tag{4.100}$$

Equivalently, for each local coefficient,

$$\boxed{L_n^m(S \to T) = \sum_{k,l} T_{n,m}^{k,l}\left(\mathbf{d}_{ST}\right) M_k^l(S).} \tag{4.101}$$

Thus, the multipole coefficients of the source cell are converted into local coefficients centered at the target cell. This operation does not require the individual source particles to be considered again: their contribution is already encoded in the multipole coefficients.

The local expansion associated with the parent cell is expressed in terms of regular solid harmonics as

$$\Phi(X) = \sum_{n=0}^{\infty}\sum_{m=-n}^{n} L_n^m \cdot Y_n^m(\theta,\phi)\cdot r^n = \sum_{n=0}^{\infty}\sum_{m=-n}^{n} L_n^m \cdot R_n^m(X) \tag{4.102}$$

where

$$L_n^m = \sum_{j=0}^{\infty}\sum_{k=-j}^{j} \frac{O_j^k \cdot i^{|m-k|-|m|-|k|} \cdot A_j^k \cdot A_n^m \cdot Y_{n+j}^{k-m}(\alpha,\beta)}{(-1)^j A_{n+j}^{k-m}\cdot \rho^{n+j+1}}, \tag{4.103}$$

with A_j^k defined by (4.91).

For each target cell T, the M2L operation is performed for all source cells S belonging to its interaction list $\mathcal{I}(T)$. The resulting contributions are accumulated into the local expansion of the target cell,

$$\boxed{\mathbf{L}^{(T)} = \sum_{S\in\mathcal{I}(T)} \mathbf{T}_{\mathrm{M2L}}\left(\mathbf{c}_T - \mathbf{c}_S\right)\mathbf{M}^{(S)}} \tag{4.104}$$

The resulting local expansion represents the far-field contribution of the source cells in the interaction list within the target cell. It is subsequently combined with the local expansion inherited from the parent cell during the local-to-local (L2L) stage.

4.2.4. Local-to-Local Translation

During the downward pass of the FMM, the local expansion of a parent cell is translated to the center of each of its children. Let P denote a parent cell and C one of its children, with centers $\mathbf{c}_P$ and $\mathbf{c}_C$, respectively. We define the translation vector as

$$\boxed{\mathbf{d}_{PC} = \mathbf{c}_C - \mathbf{c}_P}. \tag{4.105}$$

Using

$$\mathbf{x} - \mathbf{c}_P = (\mathbf{x} - \mathbf{c}_C) + \mathbf{d}_{PC}, \tag{4.106}$$

the addition theorem for regular solid harmonics allows the local expansion to be re-expressed with respect to the child center. The resulting local coefficients are obtained through the local-to-local translation operator,

$$\boxed{\mathbf{L}^{(P\to C)} = T_{LL}\left(\mathbf{d}_{PC}\right)\mathbf{L}^{(P)}.} \tag{4.107}$$

In terms of the individual coefficients, this operation can be written as

$$\boxed{L_n^m(P \to C) = \sum_{k,l} T_{LL,n,m}^{k,l}\left(\mathbf{d}_{PC}\right) L_k^l(P).} \tag{4.108}$$

The coefficients of the translation operator T_{LL} are expressed in terms of the coefficients S_n^m associated with the translation of the regular solid harmonics. In particular, the translation coefficients have the generic form

$$\boxed{T_{LLn,m}^{k,l}\left(\mathbf{d}_{PC}\right) = S_{n-k}^{,m-l}\left(\mathbf{d}_{PC}\right),} \tag{4.109}$$

with the precise index ranges and normalization determined by the chosen definition of the solid harmonics.

Therefore, the L2L operation can be expressed as

$$\boxed{L_n^m(P \to C) = \sum_{k=0}^{n} \sum_{l=-k}^{k} S_{n-k}^{,m-l}\left(\mathbf{d}_{PC}\right) L_k^l(P).} \tag{4.110}$$

If the child already contains a local contribution obtained directly from the M2L interactions, the translated contribution inherited from the parent is added to it,

$$\boxed{L_n^m(C) = L_{n\,\mathrm{M2L}}^m(C) + L_n^m(P \to C).} \tag{4.111}$$

Thus, the L2L stage propagates the local expansion down the tree, while preserving its representation in the regular solid-harmonic basis. The coefficients S_n^m enter only through the translation operator and do not constitute the basis of the local expansion.

Here is the formulation of the spherical harmonic expansion and its corresponding transformation coefficients, the field potential $\Phi(X)$ is represented as a series expansion using spherical harmonics:

$$\Phi(X) = \sum_{n=0}^{p} \sum_{m=-n}^{n} L_n^m \cdot Y_n^m(\theta, \phi) \cdot r^n, \tag{4.112}$$

where the coefficients L_n^m are defined as:

$$L_n^m = \sum_{j=n}^{p} \sum_{k=-j}^{j} \frac{O_j^k \cdot i^{|k|-|k-m|-|m|} \cdot A_{j-n}^{k-m} \cdot A_n^m \cdot Y_{j-n}^{k-m}(\alpha, \beta) \cdot \rho^{j-n}}{(-1)^{j+n} \cdot A_j^k}, \tag{4.113}$$

5 | Particle Dynamics

5.1. From Potential to Force

Once the potential has been evaluated at the target particle positions, the resulting field can be used to determine the force acting on each particle. For a conservative interaction, the force is obtained as the negative spatial gradient of the potential,

$$\boxed{\mathbf{F}_i = -\nabla\Phi(\mathbf{x}_i)}. \tag{5.1}$$

Therefore, the FMM provides the potential required to evaluate the interaction force at each particle position. In particular, the far-field contribution is obtained through the multipole and local expansions, while the near-field contribution is computed directly, as described in the previous chapters.

The force can consequently be decomposed according to the same near-field and far-field contributions,

$$\mathbf{F}_i = \mathbf{F}_i^{\text{near}} + \mathbf{F}_i^{\text{far}}, \tag{5.2}$$

where

$$\mathbf{F}_i^{\text{near}} = -\nabla\Phi_i^{\text{near}}, \qquad \mathbf{F}_i^{\text{far}} = -\nabla\Phi_i^{\text{far}}. \tag{5.3}$$

5.2. Equation of Motion

Once the force acting on each particle has been determined, its motion is governed by the corresponding equation of motion. Under a Newtonian formulation, the position $\mathbf{x}_i(t)$ of particle i satisfies

$$\boxed{m_i \frac{d^2\mathbf{x}_i}{dt^2} = \mathbf{F}_i}. \tag{5.4}$$

Introducing the particle velocity

$$\mathbf{v}_i = \frac{d\mathbf{x}_i}{dt}, \tag{5.5}$$

the second-order equation can equivalently be written as the first-order system

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i, \tag{5.6}$$

$$\frac{d\mathbf{v}_i}{dt} = \frac{\mathbf{F}_i}{m_i}. \tag{5.7}$$

Thus, at each time step, the FMM is used to evaluate the interaction field and the corresponding force, which is then used to advance the particle positions and velocities.

5.3. Time Integration

The equations of motion are integrated in time to obtain the trajectory of each particle. Let t^n denote the current time and Δt the time step. Given the particle positions and velocities at time t^n, the force is first evaluated using the FMM. The chosen time-integration scheme is then used to obtain the state of the particles at the next time level,

$$(\mathbf{x}_i^n, \mathbf{v}_i^n) \longrightarrow \left(\mathbf{x}_i^{n+1}, \mathbf{v}_i^{n+1}\right). \tag{5.8}$$

The procedure is repeated for each time step, with the FMM being re-evaluated as the particle positions evolve.

5.4. Particle-Dynamics Workflow

The complete particle-dynamics procedure can therefore be summarized as

$$\boxed{\mathbf{x}^n \xrightarrow{\text{FMM}} \Phi^n \xrightarrow{-\nabla} \mathbf{F}^n \xrightarrow{\text{Time Integration}} \left(\mathbf{x}^{n+1}, \mathbf{v}^{n+1}\right)}. \tag{5.9}$$

The updated particle positions are then used to construct the interaction problem at the next time step. The FMM tree and its associated expansions are consequently updated according to the new particle configuration.

5.5. Examples

5.5.1. Electrostatic Interaction

The particle dynamics can be illustrated by considering the electrostatic interaction between charged particles.

2D dimension

The fundamental Green's function kernel in 2D, defined as:

$$G(z_i, z_j) = \log|z_i - z_j| \tag{5.10}$$

governs the interaction between particles in both the *near-field* and the *far-field*. The key distinction lies solely in how the kernel is evaluated in each regime.

1. **Near-Field Regime: Exact Kernel Evaluation**
In the near-field, the logarithmic kernel is evaluated exactly and directly by summing the pairwise contributions from all particles j in the immediate neighborhood $\mathcal{N}(i)$ of particle i:

$$\boxed{\phi_i^{\text{near}} = \sum_{j\in\mathcal{N}(i)} q_j \log|z_i - z_j| = \sum_{j\in\mathcal{N}(i)} q_j \frac{1}{2}\log\left((x_i - x_j)^2 + (y_i - y_j)^2\right)} \tag{5.11}$$

The corresponding near-field force contribution is obtained directly from the analytical gradient of the kernel:

$$\nabla\phi_i^{\text{near}} = \sum_{j\in\mathcal{N}(i)} q_j \begin{pmatrix} \frac{x_i - x_j}{(x_i-x_j)^2+(y_i-y_j)^2} \\ \frac{y_i - y_j}{(x_i-x_j)^2+(y_i-y_j)^2} \end{pmatrix} \tag{5.12}$$

2. **Far-Field Regime**
In the far-field, to avoid an $\mathcal{O}(N^2)$ computational complexity, the exact kernel $\log(z - z_j)$ is not evaluated pairwise. Instead, it is **expanded into a series** around the source cell center z_S:

$$\log(z - z_j) = \log(z - z_S) + \log\left(1 - \frac{z_j - z_S}{z - z_S}\right) = \log(z - z_S) - \sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{z_j - z_S}{z - z_S}\right)^n \tag{5.13}$$

Through multipole-to-local (M2L) transformations, this series expansion of the kernel is converted into a **local expansion** centered at the target cell center z_T:

$$\boxed{\phi_i^{\text{far}} = \text{Re}\left[\sum_{n=0}^{p} L_n^{(T)}(z_i - z_T)^n\right]} \tag{5.14}$$

Here, the complex coefficients $L_n^{(T)}$ encode the aggregated effect of distant source particles, derived directly from the truncated series expansion of the logarithmic kernel.

The far-field force component is calculated using the complex derivative of the local expansion:

$$\frac{d}{dz}\left[\sum_{n=0}^{p} L_n^{(T)}(z - z_T)^n\right] = \sum_{n=1}^{p} nL_n^{(T)}(z - z_T)^{n-1}$$

$$\nabla\phi_i^{\text{far}} = \begin{pmatrix} \operatorname{Re}\left[\sum_{n=1}^{p} nL_n^{(T)}(z_i - z_T)^{n-1}\right] \\ -\operatorname{Im}\left[\sum_{n=1}^{p} nL_n^{(T)}(z_i - z_T)^{n-1}\right] \end{pmatrix} \tag{5.15}$$

3D dimension

In the three-dimensional formulation, the FMM was introduced using the normalized kernel

$$G(\mathbf{x}, \mathbf{x}_j) = \frac{1}{|\mathbf{x} - \mathbf{x}_j|}. \tag{5.16}$$

For an electrostatic system, the physical potential generated by a set of point charges Q_j in free space is given by Coulomb's law,

$$\boxed{\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \sum_j \frac{Q_j}{|\mathbf{x} - \mathbf{x}_j|}}, \tag{5.17}$$

where ε_0 is the vacuum permittivity.

Comparing this expression with the kernel used by the FMM, the physical constant can be absorbed into the source strength. In particular, defining

$$\boxed{q_j = \frac{Q_j}{4\pi\varepsilon_0}}, \tag{5.18}$$

the electrostatic potential takes exactly the same form used in the FMM formulation,

$$\Phi(\mathbf{x}) = \sum_j q_j G(\mathbf{x}, \mathbf{x}_j). \tag{5.19}$$

Thus, the FMM itself is independent of the physical normalization of the interaction. The algorithm computes the contribution generated by the source strengths q_j, while the physical interpretation of these coefficients depends on the interaction model being considered.

For the electrostatic case, the force acting on a target particle is obtained from the gradient of the potential,

$$\boxed{\mathbf{F}_i = -Q_i \nabla\Phi(\mathbf{x}_i).} \tag{5.20}$$

Therefore, once the FMM has evaluated the potential and its spatial gradient at the particle positions, the resulting forces can be used to advance the particle positions according to the chosen time integration scheme.

The explicit calculations are presented below.

The total potential is expressed by

$$\Phi(X) = \Phi_{\text{near}}(X) + \Phi_{\text{far}}(X) \tag{5.21}$$

where the far contribution is obtained through the multipole and local expansion truncated at order p, and the near one through the Coulomb rule:

$$\boxed{\Phi(x_i) = \sum_{q \in Ni} \frac{q}{\|x_i - x_q\|} + \sum_{n=0}^{p} \sum_{m=-n}^{n} L_n^m \cdot Y_n^m(\theta, \phi) \cdot r^n} \tag{5.22}$$

For the first part can be defined $\vec{r}_{iq} = \vec{x}_i - \vec{x}_q = (x_i - x_q, y_i - y_q, z_i - z_q)$ and $r_{iq} = \|\vec{r}_{iq}\|$. Then:

$$\Phi_{\text{near}}(x_i) = \sum_{q \in Ni} \frac{q_q}{r_{iq}} = \sum_j q_q \left((x_i - x_q)^2 + (y_i - y_q)^2 + (z_i - z_q)^2\right)^{-1/2}$$

The derivative with respect to x_i (and similarly for y_i, z_i)

$$\frac{\partial}{\partial x_i}\left[r_{iq}^{-1}\right] = -\frac{1}{2} r_{iq}^{-3} \cdot 2(x_i - x_q) = -\frac{x_i - x_q}{r_{iq}^3}$$

Therefore:

$$\frac{\partial \Phi_{\text{near}}}{\partial x_i} = -\sum_q q_q \frac{x_i - x_q}{r_{iq}^3}$$

For the second part the spherical harmonics are denoted by:

$$Y_n^m = \sqrt{\frac{(n-|m|)!}{(n+|m|)!}}\, P_n^{|m|}(\cos\theta)\, e^{im\phi}$$

where

$$P_n^{|m|}(\cos\theta) = (-1)^{|m|}\left(1 - \cos^2\theta\right)^{|m|/2} \frac{d^{|m|}}{dx^{|m|}} P_n(x) \tag{5.23}$$

The gradient of the far component is given by:

$$\begin{aligned}
\nabla \Phi_{far}(\theta, \phi, r) &= \sum_{n=0}^{p} \sum_{m=-n}^{n} \nabla\left(L_n^m Y_n^m(\theta, \phi)\right) r^n \\
&= \sum_{n=0}^{p} \sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} \nabla\left(P_n^{|m|}(\cos\theta) e^{im\phi} r^n\right)
\end{aligned}$$

The derivative with respect to θ is:

$$\begin{aligned}\frac{\partial\Phi}{\partial\theta} &= \sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} r^n \frac{\partial}{\partial\theta}\left(P_n^{|m|}(\cos\theta)\right) e^{im\phi} \\ &= \sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} r^n \left(\frac{n\cos\theta P_n^{|m|}(\cos\theta) - (n+|m|)P_{n-1}^{|m|}(\cos\theta)}{|\sin\theta|}\right) e^{im\phi}\end{aligned}$$

The derivative with respect to ϕ is:

$$\frac{\partial\Phi}{\partial\phi} = \sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} r^n \cdot P_n^{|m|}(\cos\theta) im e^{im\phi}$$

The derivative with respect to r is:

$$\frac{\partial\Phi}{\partial r} = \sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} n r^{n-1} P_n^{|m|}(\cos\theta) e^{im\phi}$$

After computing the analytical expressions for the individual partial derivatives of the potential $\Phi(r,\theta,\phi)$, we can assemble the complete gradient vector in spherical coordinates. Recall that the gradient operator in a spherical orthonormal basis $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ is defined as:

$$\nabla\Phi = \frac{\partial\Phi}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial\Phi}{\partial\theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial\Phi}{\partial\phi}\hat{\boldsymbol{\phi}} \tag{5.24}$$

To compute the physical force vector $\mathbf{F}$ from the scalar potential $\Phi(r,\theta,\phi)$, we evaluate the negative gradient of the potential, defined as $\mathbf{F} = -\nabla\Phi$. In a spherical orthonormal basis $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$, the force components are structured as follows:

$$\mathbf{F} = -\left(\frac{\partial\Phi}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial\Phi}{\partial\theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial\Phi}{\partial\phi}\hat{\boldsymbol{\phi}}\right) \tag{5.25}$$

By substituting the previously derived partial derivatives and incorporating the negative sign, the spherical force components (F_r, F_θ, F_ϕ) are explicitly given by:

$$\begin{aligned}F_r &= -\frac{\partial\Phi}{\partial r} = -\sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} n r^{n-1} P_n^{|m|}(\cos\theta) e^{im\phi} \\ F_\theta &= -\frac{1}{r}\frac{\partial\Phi}{\partial\theta} = -\sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} r^{n-1} \left(\frac{n\cos\theta P_n^{|m|}(\cos\theta) - (n+|m|)P_{n-1}^{|m|}(\cos\theta)}{\sin\theta}\right) e^{im\phi} \\ F_\phi &= -\frac{1}{r\sin\theta}\frac{\partial\Phi}{\partial\phi} = -\sum_{n=0}^{p}\sum_{m=-n}^{n} L_n^m \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} r^{n-1} \frac{P_n^{|m|}(\cos\theta)}{\sin\theta} im e^{im\phi}\end{aligned}$$

To transform this force system from the local spherical basis to the global Cartesian reference frame (F_x, F_y, F_z), we apply the standard orthogonal rotation matrix. The algebraic mapping is performed via the following matrix multiplication:

$$\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} F_r \\ F_\theta \\ F_\phi \end{bmatrix} \tag{5.26}$$

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